

MUHWEZI MARK

Telecommunications Engineer | Network Infrastructure and Backend Systems

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<https://github.com/muhwezim78> | <https://muhwezi-mark-portfolio.web.app> | <https://terraconnect-uganda.web.app> | <https://quicknetwifi.com>

PROFILE

Telecommunications engineer working on ISP infrastructure and the software that runs it. I currently provision FTTH subscribers and handle escalations as a Level 1 NOC Engineer at Savanna Fibre, and I have three years of field experience installing and troubleshooting Cambium microwave links. Alongside that I built and run TerraConnect Uganda, a multi-tenant billing and authentication platform used by hotspot operators in Uganda and QuickNet Router Management and Billing System for hotspot users in Nigeria. The work covers radio alignment and RouterOS configuration as much as it does Rust services, PostgreSQL and mobile money integrations. Degree completed at Kyambogo University, awaiting final results.

CORE SKILLS

- **Networking** Cambium ePMP and PTP 550 microwave links, cnMaestro management, MikroTik RouterOS (hotspot, PPPoE, PPP authentication), FreeRADIUS, WireGuard, IP addressing and subnetting, CGNAT mitigation, link optimisation, fault diagnosis
- **Backend and Data** Rust, Node.js, PostgreSQL, REST API design, Docker, Ubuntu server administration
- **Web and Mobile** React.js, HTML and CSS, native Android (Java, Kotlin), Flutter cross-platform development
- **Cloud, Payments and APIs** Firebase (Auth, Firestore, Cloud Storage, Hosting, Cloud Messaging), AWS, DNS and domain configuration, mobile money through pawaPay (MTN MoMo and Airtel Money), Africa's Talking (SMS and USSD), Resend transactional email, push notification delivery
- **Voice and VoIP** Asterisk PBX deployment and configuration, SIP extensions and trunking, softphone provisioning (Zoiper), Grandstream UCM intercom system design
- **Security** Network threat modelling with Security Onion, Nmap, WireGuard
- **Field and Design Tools** TEMS Investigation, walk test and drive test, Cisco Packet Tracer, AutoCAD, SolidWorks, EdrawMax
- **Embedded** ESP32, Raspberry Pi, STM32

EXPERIENCE

Founder and Lead Engineer | Blossom Tech Solutions / TerraConnect Uganda, Kampala 2023 to present

- Built and run TerraConnect Uganda, a B2B platform that gives ISP and WiFi hotspot operators billing, subscriber management, captive portals, voucher management, and mobile money collections and payouts.
- Backend is Rust and Node.js on PostgreSQL, deployed as Docker services on a production Ubuntu server. FreeRADIUS authenticates MikroTik hotspot routers at each site over WireGuard tunnels, which keeps authentication and accounting centralised.
- Integrated PawaPay for MTN MoMo and Airtel Money transactions, and took the platform through a multi-round payment provider due diligence covering AML/CFT policy, operator agreements and audit trails.
- Seven live hotspot businesses across Kampala run on the platform. I handle site survey, router provisioning, operator training and billing support for each of them.
- Delivered a router management and billing system, plus a Flutter cross-platform application, for QuickNet Networks in Nigeria (quicknetwifi.com).
- Deployed an Asterisk PBX on AWS with SIP extensions and Zoiper softphone provisioning, and designed a Grandstream UCM intercom system for on-site voice.
- Published a native Android music player to Google Play and built client web platforms hosted on AWS.

FTTH Provisioning Engineer | Savanna Fibre Limited, Kampala 2026 to present

- Provision fibre-to-the-home subscribers for a live Kampala ISP: installation, ONT configuration, activation and fault resolution at the customer premises.

PMP Field Engineer | Japotech Solutions Uganda Limited, contracted to Liquid Intelligent Technologies 2023 to present

- Field engineering on the Liquid Intelligent Technologies network as a Japotech contractor. Joined as an intern and was recruited into the role.
- Installed 180 client links on Cambium ePMP and PTP 550 point-to-point and point-to-multipoint systems during 2025, covering site survey, mounting, alignment and commissioning. Backed by Cambium vendor training in installation, link optimisation and cnMaestro management.
- Resolved 219 microwave link fault escalations. The faults ranged across interference, alignment drift, power and configuration.
- Carried out telecom site installations, walk tests and drive tests, and produced network survey documentation for the projects team.
- Awarded the company Certificate of Excellence for Integrity in 2024.

Intern, Field Maintenance and Transmission | Innovis Telecom Services International Ltd *May to July 2025*

- Six-week structured rotation based at the Airtel Uganda MSC in Bugolobi, covering field maintenance, transmission, data centre maintenance, network optimisation and drive test, and health and safety compliance.

EDUCATION

BSc Telecommunications Engineering | Kyambogo University *2022 to 2026, awaiting results*

- Final year project: TerraConnect Uganda, an automated router management system addressing the limitations of CGNAT. Supervised by Dr. Mugerwa Dickson.

Uganda Advanced Certificate of Education, Standard College Ntungamo, 2020. Principal passes in Mathematics, Physics and Chemistry.

Uganda Certificate of Education, Kitante Hill School, 2018. Division 1, ten subjects passed.

CERTIFICATIONS AND TRAINING

Cambium Networks Learning vendor training completed 2024

ePMP track: Installation Training for Force 180, 190, 200 and 300-25; Access Point Device Configuration; Link Optimization; Force 300-25 solution training.

PTP 550 track: Solution Overview; Configuring PTP 550; Configuring Antenna Gain; Understanding Metrics; Remote Management with cnMaestro.

Other

Internship Programme Completion Certificate, Innovis Telecom Services International Ltd, 2025.

Certificate of Excellence, Company Value of Integrity, Japotech Solutions, 2024.

Telecom site installation, point-to-multipoint installation, walk test and drive test, Japotech Solutions.

Website development, hosting and native Android development, Blossom Tech Solutions.

OPEN SOURCE

1,134 contributions over the past year across 55+ repositories, including network backends, MikroTik integration tooling and trading systems. github.com/muhwezim78

LANGUAGES

English, Luganda and Runyankore. Fluent in all three.

REFEREES

Mr. Jamil Wamala, Projects Team Leader, Japotech Solutions. +256 787 157 267

Mrs. Lyn Kyimezire, Assistant Projects Team Leader, Japotech Solutions. +256 757 052 015

Mr. Nyero Kenneth, Senior Developer, Blossom Tech Solutions. +256 758 830 464

Mr. Lutswala Emma, Field Engineer, Innovis Telecom Services International. +256 700 554 928